

# **SafeLAN: SECURE FILE SYSTEM FOR LAN**

Project Id: 25-26J-067

Project Proposal Report

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BSc. Special (Hons.) Degree in Information Technology

(Specialized in Cyber Security)

Department of Computer System Engineering

Sri Lanka Institute of Information Technology

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## DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.



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## ABSTRACT

The increasing need for secure file access within local area networks (LANs) has highlighted the limitations of traditional authentication methods, such as static passwords or operating system level permissions, which cannot verify user identity continuously or detect anomalous behavior. This study proposes the development of a dynamic access control system for LAN-based file systems using behavioural biometrics, specifically keystroke dynamics data. This component aligns directly with cybersecurity principles of authentication and access control. Upon user login, the system continuously monitors user activity, calculates trust scores based on behavioral profiles and contextual attributes like user role, device, and time of access and dynamically enforces access control by granting or denying permissions in real time. By integrating machine learning techniques for behavioral analysis, this approach enhances session security, reduces insider threats, and strengthens identity verification beyond static authentication methods. The study also involves the collecting and analysis of needs, which might include both functional and non-functional requirements, as well as elements of commercializing the tool.

*Key Words - dynamic access control, behavioral biometrics, trust score, continuous authentication, LAN security*

## TABLE OF CONTENTS

|                                           |    |
|-------------------------------------------|----|
| DECLARATION .....                         | 1  |
| ABSTRACT .....                            | 2  |
| TABLE OF CONTENTS .....                   | 3  |
| LIST OF TABLES .....                      | 4  |
| LIST OF FIGURES .....                     | 4  |
| 1 INTRODUCTION .....                      | 5  |
| 1.1 Background .....                      | 5  |
| 1.2 Literature Survey .....               | 7  |
| 1.3 Research Gap .....                    | 8  |
| 1.4 Research Problem .....                | 9  |
| 2 OBJECTIVES .....                        | 10 |
| 2.1 Main Objectives .....                 | 10 |
| 2.2 Specific Objectives .....             | 10 |
| 3 METHODOLOGY .....                       | 11 |
| 3.1 System Overview and Diagram .....     | 11 |
| 3.2 Datasets .....                        | 13 |
| 3.3 Gantt Chart .....                     | 14 |
| 4 REQUIREMENTS .....                      | 15 |
| 4.1 Functional Requirements .....         | 15 |
| 4.2 Non-Functional Requirements .....     | 16 |
| 5 BUDGET AND BUDGET JUSTIFICATION .....   | 17 |
| 5.1 Data Collection .....                 | 17 |
| 5.2 Development Tools and Libraries ..... | 17 |
| 5.3 System Design .....                   | 17 |

|       |                                           |    |
|-------|-------------------------------------------|----|
| 5.4   | Integrated Development Environment .....  | 17 |
| 5.5   | Version Control.....                      | 17 |
| 5.6   | Documentation .....                       | 18 |
| 6     | COMMRCIALIZATION .....                    | 19 |
| 6.1   | Market Opportunities .....                | 19 |
| 6.1.1 | Government Sector.....                    | 19 |
| 6.1.2 | Healthcare Sector .....                   | 19 |
| 6.1.3 | Banking and Financial Sector .....        | 19 |
| 6.1.4 | Research and Education Institutions ..... | 19 |
| 6.1.5 | Corporate Enterprises.....                | 20 |
| 7     | REFERENCE LIST .....                      | 21 |

## **LIST OF TABLES**

|           |                   |   |
|-----------|-------------------|---|
| Table 1.1 | Research Gap..... | 8 |
|-----------|-------------------|---|

## **LIST OF FIGURES**

|            |                                           |    |
|------------|-------------------------------------------|----|
| Figure 3.1 | Authentication Module System Diagram..... | 11 |
| Figure 3.2 | Flow diagram .....                        | 12 |
| Figure 3.3 | Gantt chart related to the component..... | 14 |

# 1 INTRODUCTION

The security of Local Area Networks (LANs) has become an essential requirement for modern organizations that handle sensitive information and rely on file-sharing systems for daily operations. While there are efficient LAN-based file sharing without requiring internet connectivity, their system lacks integrated security mechanisms, which are key components of the proposed LAN-based file system, referred to in this study as SafeLAN [1]. Traditional authentication methods, such as static passwords, operating system-level permissions, or basic encryption, provide only limited protection because they only focus on verifying whether credentials are correct at the time of login. At this stage, these traditional methods fail to provide strong protection as they cannot ensure that the individual entering the credentials is the legitimate account owner. This creates opportunities for credential misuse, unauthorized access, and insider threats. This study focuses on using keystroke dynamics of credential input as a behavioral biometric to provide continuous, adaptive authentication and enhance security in the LAN-based file system.

## 1.1 Background

The SafeLAN system provides a LAN-only secure file-sharing environment that integrates continuous behavioural authentication, intelligent monitoring, and tamper-evident audit logging. It ensures data confidentiality, integrity, and accountability without relying on default OS-level permissions, making it suitable for small to medium enterprises (SMEs) seeking a robust, offline file security solution.

The “Behavioral Biometrics–Based Adaptive Authentication Module” enhances the security of the LAN-based file system, by continuously validating users through their unique interaction patterns. Instead of relying solely on static credentials, this module analyzes behavioural patterns such as typing patterns and typing speed during login to authenticate legitimate users. By comparing observed behaviours against a trained model, the module can detect anomalies, prevent unauthorized access, and mitigate insider threats.

To address this limitation, this study shows how behavioral biometrics can be used to strengthen authentication at login. In particular, keystroke dynamics enable the system to analyze patterns in user behavior, making it possible to accurately identify users and verify their identity before granting access to the system. This approach enhances the traditional login process by adding a layer of adaptive security based on individual behavior, reducing the risk of unauthorized access even when static credentials are compromised.

The system calculates a trust score based on behavioural profiles and contextual attributes to determine whether the user should be granted access. Machine learning techniques are employed to analyse keystroke data, ensuring accurate verification of user identity at the point of system login. The proposed system aligns with cybersecurity principles of authentication and access control, offering a secure and reliable login mechanism for LAN-based file systems.



## 1.2 Literature Survey

The increasing reliance on Local Area Networks (LANs) for sharing sensitive information has highlighted the limitations of static authentication methods. Traditional authentication systems verify credentials only at login, failing to detect abnormal usage by legitimate users by leaving systems vulnerable to insider threats and data breaches. Behavioral biometrics, specifically keystroke dynamics, provide a solution by capturing unique user interaction patterns. Keystroke features such as dwell time, flight time, and typing rhythm have been widely studied and shown effective for authentication [2], [3], [4].

Early studies employed classical machine learning methods, including Support Vector Machines (SVMs) and Random Forests, to classify user behavior with moderate success [5]. More recent approaches leverage deep learning techniques such as CNNs [6] and RNNs (LSTM, GRU) [7], [8] to capture temporal typing patterns more effectively by improving authentication accuracy [9], [10] [11]. Datasets such as the CMU and Kaggle keystroke benchmarks [12], [13] have enabled comparative evaluations and helped refine model performance.

Surveys and analyses highlight the growing shift toward adaptive systems that integrate multiple behavioral features [3], [14]. Recent works propose trust score-based methods, where access decisions incorporate contextual attributes (user role, device, time) alongside behavioural biometrics, enabling more dynamic and resilient authentication [15] [14], [16]. This project builds on these advances by combining keystroke dynamics and contextual attributes data into a trust score based authentication framework, offering a lightweight yet adaptive authentication mechanism for LAN file systems.

### 1.3 Research Gap

Behavioural biometrics, specifically keystroke dynamics, have been extensively researched for user authentication. Previous researches have used machine learning models like SVMs and Random Forests to successfully classify users based on their typing patterns, but these studies have a number of drawbacks when it comes to LAN-based file system authentication.

The table below compares, the methods used in “Behavioral Biometrics–Based Adaptive Authentication Module for SafeLAN File System” with the previous researches.

| Research         | Behavioural Biometrics | Contextual Login | Trust Score-Based Access | Adaptive | LAN-Focused |
|------------------|------------------------|------------------|--------------------------|----------|-------------|
| Research A       | ✓                      | ✗                | ✗                        | ✗        | ✗           |
| Research B       | ✓                      | ✗                | ✗                        | ✓        | ✗           |
| Research C       | ✗                      | ✓                | ✗                        | ✗        | ✗           |
| Research D       | ✗                      | ✗                | ✗                        | ✗        | ✓           |
| Proposed Project | ✓                      | ✓                | ✓                        | ✓        | ✓           |

Table 1.1 Research Gap

## 1.4 Research Problem

Traditional authentication mechanisms in LAN-based file systems, such as static passwords or operating system level permissions authenticate users only once at login. This, leaves systems vulnerable to insider threats, data breaches, and unauthorized access after login. Current LAN environments also lack integration of behavioral biometrics, contextual login and trust-based scoring mechanisms, which are needed to ensure real-time, adaptive access control. Therefore, the problem is the absence of a secure, dynamic authentication model that can continuously validate user identity and prevent unauthorized file access within LAN systems.

The research problem can be formulated as the following questions:

- How can behavioral biometrics, such as keystroke dynamics and mouse movements, be integrated with contextual login attributes to enhance authentication in LAN-based file systems?
- Can a trust score-based adaptive authentication mechanism provide stronger resistance against insider threats compared to traditional static login methods?
- Can continuous, behavior-driven validation during login in LAN environments effectively minimize the risk of unauthorized access and credential compromise?

While behavioral biometrics such as keystroke dynamics have addressed continuous authentication, most existing researches are limited to general computing environments and not tailored for LAN-based file systems. Furthermore, existing LAN-focused file sharing solutions emphasize connectivity and efficiency but lack robust security and authentication layers [1].

Another limitation of prior studies is the absence of contextual login mechanisms, which are necessary to improve the reliability of authentication. Similarly, the lack of trust score-based adaptive access control prevents these systems from responding dynamically to changes in user behaviour or environment. This results in rigid, one-time authentication that fails to provide ongoing verification and leaves critical gaps in protecting sensitive LAN data.

## **2 OBJECTIVES**

### **2.1 Main Objectives**

Main objective is to bridge the security gap in LAN based access control systems by introducing a real-time biometric authentication layer, reducing the attack surface associated with relying solely on static passwords. The component introduces a behaviour-driven authentication layer for LAN-based file systems, replacing static, one-time verification with a more intelligent approach. By capturing and analyzing unique keystroke patterns during login, the system creates a behavioural signature that is difficult for attackers to replicate. This signature is further strengthened by a trust score mechanism, which incorporates contextual attributes such as user role, device, and time of access. The outcome is a login process that adapts to user behavior, providing stronger protection against unauthorized access and minimizing the risk of compromised credentials in LAN environments.

### **2.2 Specific Objectives**

- Analyse keystroke data from users during login to create behavioural profiles. By capturing patterns such as typing rhythm, flight time, and dwell time, the system can distinguish legitimate users from potential impostors.
- Develop a trust score mechanism that incorporates behavioural patterns and contextual attributes (e.g. user role, device, time of access).
- Implement a dynamic access control system that grants or denies access based on real-time trust score evaluation. By continuously assessing authentication at login, the system ensures that only verified users can access sensitive files within the LAN.
- Reduce reliance on static passwords by providing an additional layer of identity verification for LAN file systems.
- Enhance LAN security by minimizing insider threats and preventing unauthorized file access.

### 3 METHODOLOGY

This project proposes a LAN-based secure file system that focuses on adaptive authentication rather than static, one-time verification. Traditional authentication methods such as passwords or role-based permissions cannot sufficiently address insider misuse or evolving attack patterns. To address this, the system leverages behavioral biometrics, particularly keystroke dynamics combined with trust scoring to evaluate the legitimacy of a user's login attempt. By incorporating both behavioural and contextual parameters along with the trust score, the solution enhances resilience against unauthorized entry, strengthens user accountability, and ensures that sensitive file operations remain accessible only to trusted individuals within the LAN environment. The following diagram represents the flow of behavioural biometrics-based adaptive authentication module for SafeLAN file system.

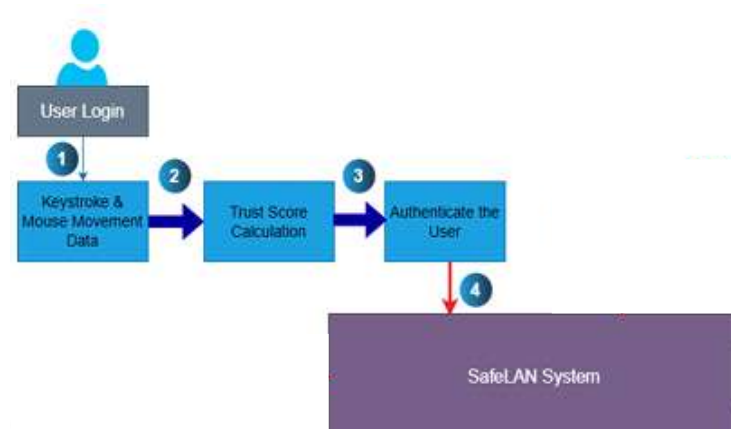


Figure 3.1 Authentication Module System Diagram

#### 3.1 System Overview and Diagram

A system overview of the keystroke dynamics model demonstrates how accuracy improves security and robustness for user authentication. It shows how to develop a hybrid model that combines Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). CNN is used to analyze the spatial data and RNN is used to analyse the temporal data. Furthermore, this includes phases such as data collection, data preprocessing, feature extraction, sequence analysis, and model integration for authentication decisions. The diagram represents the flow of these components.

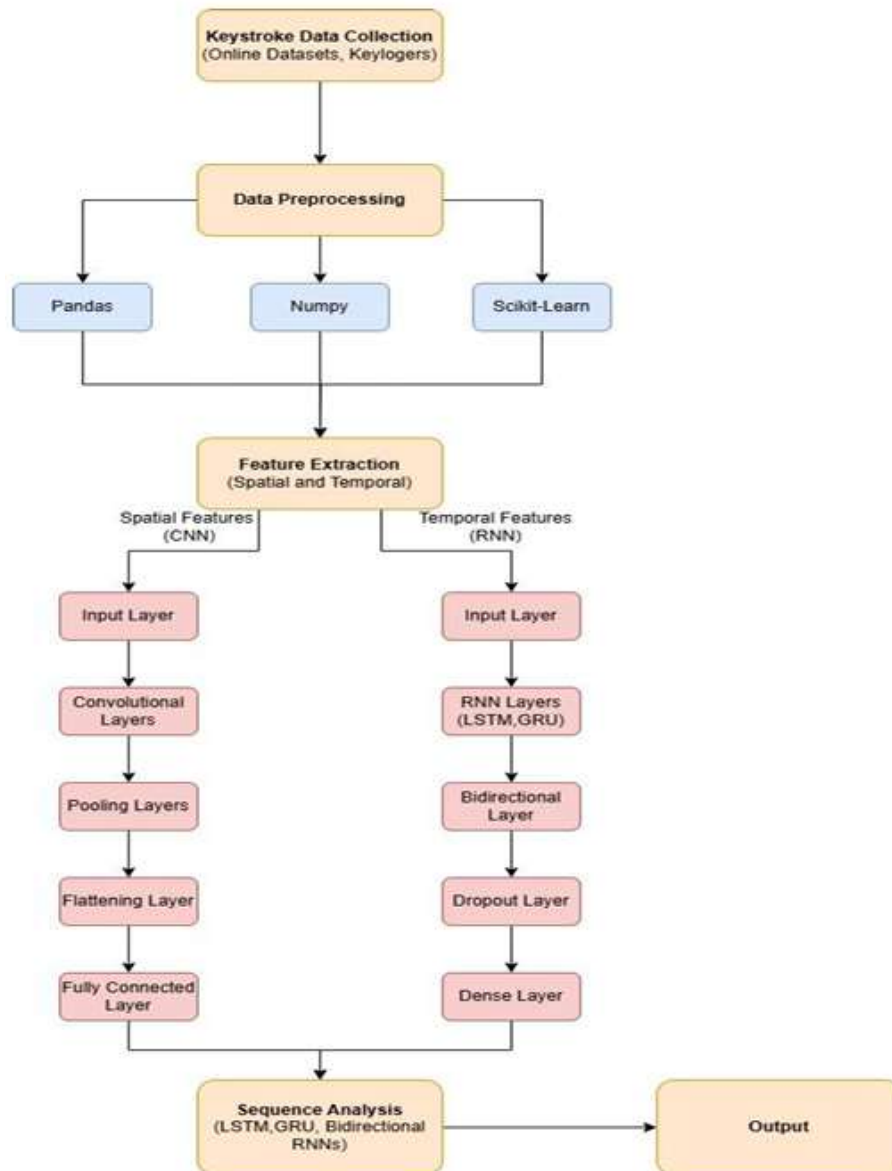


Figure 3.2 Flow diagram

The proposed methodology integrates behavioural biometrics, specifically keystroke dynamics, with a trust scoring system to continuously evaluate the authenticity of a user's actions. Behavioral data such as typing speed, key hold time, and key transition patterns are needed to develop this module. This is combined with contextual parameters like login time, location, and device usage patterns to calculate a dynamic trust score for each user.

The overall methodology includes the following conceptual steps:

1. **Data Collection:** Obtain keystroke and mouse dynamics from publicly available datasets (e.g. Kaggle) that simulate normal user behaviour. This data serves as a basis for model development and testing, allowing the system to analyse behavioural patterns without needing live user data at this stage.
2. **Data Preprocessing:** Clean and normalize the collected data to remove inconsistencies, missing values, or anomalies, ensuring that the behavioural patterns are suitable for analysis and model training.
3. **Feature Extraction:** Identify key behavioural traits such as typing rhythm, key hold duration, key transitions, mouse movements, and click patterns from the preprocessed data.
4. **Sequence Analysis:** Use temporal analysis methods to observe patterns over time. For example, recurrent neural networks (RNNs) could theoretically capture sequence dependencies in typing behavior.
5. **Model Integration:** A hybrid model combining CNN (for spatial patterns, e.g., typing distribution) and RNN (for temporal sequences, e.g., keystroke timing) can be used conceptually to evaluate user behaviour.
6. **Trust Score Computation:** Assign dynamic trust scores based on deviations from the user's normal behavioural pattern. Higher deviation means lower trust score.
7. **Adaptive Access Control:** Grant, restrict, or revoke access to files depending on the trust score and contextual risk parameters.

### 3.2 Datasets

To develop the keystroke dynamic authentication module multiple datasets like keystroke dynamics are needed.

- Public datasets like CMU Keystroke Dataset, MIT's keystroke datasets, or synthetic datasets collected from volunteers.
- Contextual information like login time, device ID, and IP address can be incorporated.
- The data can be divided into training data (normal user behaviour) and test data (to simulate anomalies or unauthorized access).

### 3.3 Gantt Chart

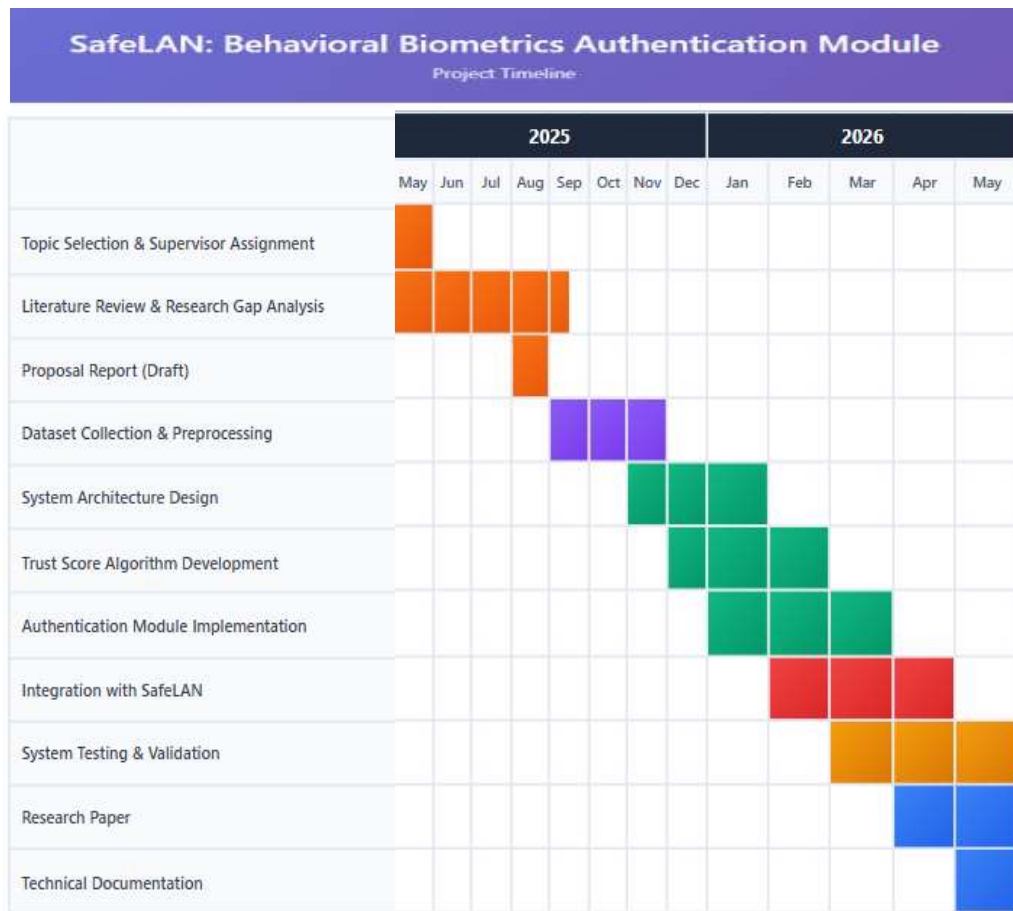


Figure 3.3 Gantt chart related to the component



## **4 REQUIREMENTS**

SafeLAN continuously authenticates users, monitors behaviour for anomalies, and enforces dynamic access control. It combines usability, cost-efficiency, and robust security measures to protect sensitive data and ensure accountability within an organization's internal network. The following requirements focus on the component-based functionality of the system, specifically the behavioural biometrics-based adaptive authentication module for the SafeLAN file system.

### **4.1 Functional Requirements**

Functional requirements describe the specific behaviours and features that the "Behavioural Biometrics-Based Adaptive Authentication Module" must provide within SafeLAN:

#### **1. User Authentication:**

- System must authenticate users based on behavioural biometrics such as keystroke dynamics and mouse movement patterns.

#### **2. Trust Score Computation:**

- System must calculate a dynamic trust score based on deviations from the user's normal behavioural patterns.
- Trust scores should be updated in real-time to reflect current user behavior.

#### **3. Adaptive Access Control:**

- System should grant, restrict, or revoke access to files based on trust scores and contextual parameters (e.g. time of access, device used).
- The system should provide alerts or requests for re-authentication if abnormal behaviour is detected.

#### **4. Data Management:**

- System must store user behavioural profiles securely for analysis and comparison.

#### **5. Reporting:**

- System should generate alerts/reports on user behaviour or anomalies detected.

These requirements ensure that the system can continuously verify user identity, compute trust scores, enforce adaptive access control, securely manage user profiles, and generate alerts or reports on anomalous behaviour.

## **4.2 Non-Functional Requirements**

Non-functional requirements describe the quality attributes and constraints of the system:

### **1. Security:**

- The system must ensure that only authorized users can access sensitive files by continuously validating behaviour patterns during each user login.

### **2. Performance:**

- Trust score computation and access control decisions should be performed with minimal delay to avoid disrupting user workflow.

### **3. Scalability:**

- System should handle multiple users simultaneously on a LAN without performance degradation.

### **4. Reliability:**

- System should operate continuously without failures and recover gracefully from unexpected interruptions.

## **5 BUDGET AND BUDGET JUSTIFICATION**

The budget for the SafeLAN File System project will cover the costs associated with developing, testing, and documenting the system, including hardware, software, data acquisition, and other necessary resources.

### **5.1 Data Collection**

- Cost of purchasing and accessing the keystroke dynamics dataset like CMU Benchmark Dataset, BKSD Dataset and more.
- Estimated Cost: \$20

However, since the behavioural data is collected and provided in dataset, keylogger implementations and related costs are fully eliminated.

### **5.2 Development Tools and Libraries**

- Includes costs for deep learning frameworks such as TensorFlow, Keras, and scikit-learn, data analysis tools, and other software required for model development and behavioural analysis.
- Estimated Cost: \$25

### **5.3 System Design**

- Covers labor, planning, and design efforts for the behavioural biometric-based authentication module in SafeLAN architecture, including GUI, modules, and access control workflows.
- Estimated Cost: \$10

### **5.4 Integrated Development Environment**

- Cost for tools and plugins for development and coding process
- Estimated Cost: \$0

### **5.5 Version Control**

- Cost for version control and collaboration tools like GitHub or Bitbucket
- Estimated Cost: \$0

## **5.6 Documentation**

- Covers report preparation, diagrams, user manuals, and technical documentation to support system deployment and future maintenance.
- Estimated Cost: \$8

## **6 COMMERCIALIZATION**

SafeLAN is a LAN-based secure file system designed for organizations that prioritize continuous authentication, and data privacy. Its commercialization strategy focuses on addressing sectors with sensitive data and high security requirements, while offering a practical, cost effective solution for small to medium enterprises (SMEs).

### **6.1 Market Opportunities**

#### **6.1.1 Government Sector**

Government agencies handle highly sensitive information, including citizen records, defense data, and classified projects. SafeLAN with behavioural biometric authentication ensures that only authorized personnel can access critical files, reducing insider threats and strengthening national security.

#### **6.1.2 Healthcare Sector**

Hospitals and medical research facilities require strict confidentiality for patient records and research data. Integrating behavioural biometrics enhances authentication beyond traditional methods, ensuring compliance with data privacy regulations such as HIPAA while preventing unauthorized access by internal staff.

#### **6.1.3 Banking and Financial Sector**

Banks and financial institutions manage large volumes of sensitive financial data. A trust score-based LAN authentication system can prevent fraud, secure internal communications, and protect customer information, offering a competitive advantage by enhancing security and regulatory compliance.

#### **6.1.4 Research and Education Institutions**

Universities and research centers often store valuable intellectual property and research data. A secure LAN system with dynamic authentication ensures only verified users access sensitive files, safeguarding proprietary information and reducing the risk of data leaks.

### **6.1.5 Corporate Enterprises**

Enterprises with internal LANs require robust security to protect strategic documents and proprietary business information. SafeLAN's behavioural biometrics and trust scoring can integrate with existing IT infrastructure to offer real-time protection against insider threats, ensuring secure file sharing and controlled access across the organization.

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